

Question Number	Answer	Additional guidance	Mark
7(a)(i)	An answer that makes reference to one of the following points: • adhesion to the skin (1) • prevent {bacteria drying out / dessication} (on the skin) (1) • {prevents / protection from} {digestion from enzymes / detection by immune system / phagocytosis / antibodies / antibiotics / chemicals} (1)	ignore protection unqualified	(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	An answer that makes reference to one of the following points: • does not live in a liquid environment (1) • does not need to move around the skin (1)	ignore it {is transported by the blood / enters a cut}	(1)

Question Number	Answer	Mark
7(a)(iii)	The only correct answer is D 1, 2, and 3 <i>A is not correct because bacterial cells also have ribosomes</i> <i>B is not correct because bacterial cells also have cell membrane</i> <i>C is not correct because bacterial cells also have a cell wall</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(i)	An answer that includes the following points: • $\sum(x - \bar{x})^2$ correctly calculated (1) • (n-1) correctly calculated (1) • correct calculation of SD (1)	<u>Example of calculation</u> $\sqrt{(6.5 \div 2)} = 1.8027$ ecf applies for mp3 only correct answer scores full marks	(3)

Question Number	Answer	Mark
7(b)(ii)	The only correct answer is A to see how widely the diameter values differ from the mean diameter <i>B is not correct because that is a chi squared test</i> <i>C is not correct because that is correlation coefficient and the student did not investigate different concentrations</i> <i>D is not correct because that is the mode</i>	(1)

Question Number	Answer	Mark
7(c)(i)	The only correct answer is A 8.3% <i>B is not correct because $(203 \div 2\,444) \times 100 = 8.3\%$</i> <i>C is not correct because $(203 \div 2\,444) \times 100 = 8.3\%$</i> <i>D is not correct because $(203 \div 2\,444) \times 100 = 8.3\%$</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	neither the {patient / group} nor the {doctor / scientist} {would know if they were taking just flucloxacillin or both flucloxacillin and clindamycin (1)	ignore references to placebo ignore references to medicine / drug / treatment unqualified	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(iii)	An answer that includes four of the following points: • {swelling / circumference difference / mean (difference)} reduces (in both / over time) (1) • greater overall reduction in {swelling / difference / mean (difference)} with group B than group A / {0.21 cm / 11.6 %} more decrease in difference for group B compared to group A (1) • treatment for group B {is more effective / has higher antimicrobial effect / has higher antibacterial effect} (1) • do not know if the circumference of leg was the same as the non-infected leg before treatment started (1) • relevant comment on study (1)	ACCEPT calculated decreases ACCEPT converse ACCEPT do not know if one antibiotic reduced the swelling faster in the first 5 days ACCEPT do not know initial {difference / circumference} e.g. sample size linked to validity of conclusions / no indication if any patients {immunosuppressed / same age} / no standard deviation so {no indication of variation in data / can't judge significance} / only males used so can't apply conclusions to females	(4)